Dissolving the Annuity Puzzle: A Critical Survey

Abstract

Yaari (1965) proved that a rational consumer facing an uncertain lifetime, with no bequest motive and access to actuarially fair annuities, should annuitize all wealth. The gap between this prediction and observed behavior – voluntary immediate annuity ownership rates of 3-6 percent across developed economies – has been labeled the "annuity puzzle." This survey argues that the aggregate puzzle, as conventionally framed for classic immediate life annuities against the Yaari benchmark, should be retired. The full-annuitization result rests on nine restrictive assumptions, none of which holds without qualification as a descriptive matter. Pre-existing annuitization places many current retirees at 50-75 percent annuitized before any voluntary purchase, depending on wealth, marital status, and defined-benefit pension coverage. Bequest motives, health expenditure risk, and stochastic mortality each substantially reduce predicted demand in calibrated models. Under prospect theory with narrow framing, predicted demand falls to zero, though the result is sensitive to reference point specification. Supply-side frictions and cross-cultural evidence reinforce this case. Multiple channels combine to predict approximately what is observed, though important questions remain – particularly regarding deferred income annuities and the heterogeneous welfare question of which households would benefit from additional annuitization.

Keywords: annuity puzzle; life-cycle theory; longevity risk; bequest motives; prospect theory; money's worth ratio

JEL Classification: D15, D81, D91, G22, H55, J26

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1. Introduction

Yaari (1965) proved that a consumer facing an uncertain lifetime, possessing no bequest motive, and with access to actuarially fair life annuities should allocate all wealth to annuity contracts. Annuities dominate conventional assets because they offer a mortality credit – a return premium financed by redistributing wealth from those who die to survivors. Davidoff, Brown, and Diamond (2005) subsequently generalized the result under weaker preference assumptions.

The empirical reality is starkly different. Voluntary life annuity purchases are vanishingly rare across developed economies. In the United States, approximately 3.6 percent of single retirees aged 65 to 69 hold privately purchased income annuities (Lockwood, 2012). In Australia, lifetime annuity products account for a negligible fraction of approximately AUD 3.6 trillion in superannuation assets (Australian Treasury, 2023). When the United Kingdom's 2015 pension freedoms removed compulsory annuitization, sales collapsed by approximately 75 percent (Financial Conduct Authority, 2018). Singapore's CPF LIFE required mandatory implementation to achieve high compliance.

Throughout this survey, "annuity demand" refers specifically to voluntary purchase of classic immediate life annuities – products converting a lump sum into an irrevocable, lifelong income stream. Deferred income annuities and hybrid products raise distinct issues discussed in Section 8. The evidence base spans distinct empirical objects – retail SPIAs, pension-plan payout choices, cross-country institutional annuitization rates – and the survey flags where conclusions are object-specific. Measurement varies across studies (ownership rates, wealth shares, flow measures), but by any measure voluntary annuitization is low relative to the Yaari benchmark.

The chasm between theory and observation was labeled the "annuity puzzle" and became one of the most durable research programs in household finance. The term itself signals an interpretive commitment: the theoretical prediction is taken as the benchmark, and observed behavior is treated as the anomaly requiring explanation. This framing has generated a vast literature in which scores of papers propose partial explanations – ranging from adverse selection and loading costs to bequest motives and cognitive limitations – each typically

modeled as an independent friction that reduces but does not eliminate optimal annuity demand. The cumulative impression left by this literature is that the puzzle has been chipped away at its margins while its core remains intact, that some residual gap persists between what rational agents should do and what households actually do.

This paper argues that the cumulative impression is overstated. Six decades of research have shown that the puzzle of low demand for classic immediate life annuities substantially dissolves once the Yaari framework's assumptions are replaced with empirically grounded alternatives. The contribution is not new data or a new model, but a demonstration that the existing literature, taken as a whole, provides multiple empirically plausible mechanisms that can rationalize the low levels of voluntary SPIA demand observed across developed economies, and that no leading framework robustly predicts high take-up once realistic institutions and preferences are incorporated. The gap between theory and evidence is primarily a gap between a fragile theoretical benchmark and the realities of human preferences, institutional constraints, and insurance markets. This dissolution is a positive finding: multiple models predict low demand under realistic parameterizations. The normative question – whether non-annuitization reduces welfare for specific subpopulations – is analytically distinct and remains open (Sections 5.6 and 8.2). The appropriate response is to retire the Yaari-benchmark puzzle framing and redirect research toward the heterogeneous welfare question: for which households, under what conditions, would additional annuitization improve welfare?

The case proceeds along several interconnected lines. Section 2 argues that the full-annuitization result rests on a conjunction of nine restrictive assumptions and shows that the result is fragile: relaxing individual assumptions in calibrated models can drive predicted demand to zero. Section 3 shows that pre-existing annuitization places the typical retiree at 50-75 percent annuitized, and that calibrated bequest motives combined with purchase frictions reduce predicted ownership to 4-5 percent, matching the observed rate. Section 4 shows how health expenditure risk raises the value of liquidity and how correlated health-mortality risk (Reichling and Smetters, 2015) can drive optimal demand to zero. Section 5 shows that under prospect theory with standard parameter values, predicted demand falls to zero, and that framing effects produce preference swings incompatible with the expected utility foundations of the Yaari result. Section 6 examines supply-side failures and cross-cultural

evidence consistent with low voluntary demand across diverse institutional contexts. Section 7 synthesizes these channels, showing that their interaction effects compound rather than merely add.

Prior reviews include Brown (2007) on the normative case for annuitization and behavioral perspectives, Benartzi, Previtero, and Thaler (2011) on behavioral and institutional factors in decumulation, and Pashchenko (2013), whose structural lifecycle model with sequential decomposition provides the most rigorous quantitative benchmark for channel-by-channel assessment. The present survey differs in integrating channels that prior surveys treated in isolation, incorporating cross-cultural evidence postdating the earlier surveys (notably the UK's 2015 pension freedoms), introducing evolutionary psychology arguments for bequest motives, and arguing explicitly for retiring the aggregate puzzle framing in favor of the heterogeneous welfare question.

Methodologically, this paper constitutes a theory-based critical survey. It does not present new empirical results or develop a novel theoretical model. Rather, it assesses the role of specific theoretical assumptions within the annuity puzzle literature, evaluates the empirical plausibility of each, and synthesizes findings from across behavioral economics, household finance, insurance economics, public economics, evolutionary psychology, and comparative institutional analysis.

The literature surveyed spans Yaari (1965) to 2025, assembled through keyword searches in EconLit, Google Scholar, RePEc/IDEAS, SSRN, and NBER Working Papers using terms including "annuity puzzle," "annuitization," "life annuity demand," "longevity insurance," and "money's worth annuity," supplemented by forward and backward citation chasing from canonical anchors (Yaari, 1965; Davidoff, Brown, and Diamond, 2005; Mitchell et al., 1999; Lockwood, 2012; Brown, 2007). The survey is selective rather than exhaustive: studies were included based on their contribution to the nine Yaari benchmark assumptions (Section 2) and to the principal channels of dissolution. Priority was given to published journal articles and working papers from recognized series (NBER, CRR); institutional reports were included for empirical benchmarks not available in academic publications. Studies reaching contrary conclusions – notably Pashchenko (2013) and Peijnenburg et al. (2016) – are discussed in Section 7.1 and engaged with throughout. Inkmann, Lopes, and Michaelides (2011), whose

calibrated model finds that reasonable preference parameters can match observed low demand, provide independent corroboration through a different modeling framework. Cross-cultural evidence is drawn from six institutional contexts (UK, US, Australia, Singapore, Netherlands, Chile).

2. The Yaari Framework: Assumptions, Extensions, and Fragility

The entire annuity puzzle rests on a single theoretical result. Yaari (1965) proved that under specific conditions, a rational consumer facing an uncertain lifespan should convert all wealth into actuarially fair life annuities. This section argues that the full-annuitization result rests on restrictive assumptions that, when replaced with empirically grounded alternatives, markedly reduce or eliminate predicted annuity demand.

2.1 The Original Yaari (1965) Model

Yaari (1965) considers a consumer who lives from time zero to a random death time T, distributed according to a known survival function, and who maximizes expected discounted lifetime utility over an uncertain horizon. Without annuities, the consumer holds conventional bonds earning a riskless rate r, and the Euler equation governing optimal consumption growth contains the mortality hazard rate as an additional discount factor. The consumer is penalized for the risk of dying with unspent resources.

With actuarially fair annuities, the picture changes dramatically. The annuity pays a return equal to the bond rate plus the mortality hazard rate – the mortality credit – financed by redistributing wealth from those who die to survivors. This credit exactly offsets the mortality discount in the Euler equation, so the consumer's optimal consumption path closely parallels that of a consumer with a certain lifetime – though Yaari (1965, pp. 146-147) notes the equivalence is "not quite true" because the solutions need not coincide in all cases. Since the annuity return strictly exceeds the bond return wherever the mortality hazard is positive, annuities stochastically dominate bonds for any consumer who derives no utility from wealth at death. Full annuitization is optimal.

The result explicitly assumes: (i) expected utility maximization; (ii) additively separable

utility; (iii) no bequest motive; (iv) actuarially fair pricing; and (v) a known, fixed survival function. Yaari's formulation uses a general subjective discount function, though exponential discounting is standard in subsequent implementations and has been treated as part of the benchmark in the applied literature (Laibson, 1997). None of these assumptions has survived empirical scrutiny unchallenged. Expected utility is contradicted by loss aversion and probability distortion (Kahneman and Tversky, 1979). Additive separability is rejected by evidence of habit formation (Constantinides, 1990) and state-dependent marginal utility (Finkelstein, Luttmer, and Notowidigdo, 2013). Approximately 75 percent of elderly singles are estimated to have bequest motives based on structural estimation of saving behavior (Kopczuk and Lupton, 2007). Annuities offer only 80-85 cents of expected payouts per premium dollar (Mitchell et al., 1999). Survival functions shift stochastically with health shocks (Reichling and Smetters, 2015). And present bias challenges exponential discounting (Laibson, 1997).

2.2 The Nine Assumptions and Their Empirical Status

Yaari's explicit assumptions do not exhaust the assumptions embedded in the full-annuitization benchmark. His model embeds additional assumptions that subsequent researchers identified as independently important. First, Yaari's consumer receives an earnings stream and holds no annuity contracts from other sources – an assumption of no pre-existing annuitized income. In reality, Social Security and DB pensions annuitize 50-75 percent of median retiree wealth (Dushi and Webb, 2004; Poterba, Venti, and Wise, 2011). Second, the only source of uncertainty is the date of death; the consumer faces no health expenditure shocks. De Nardi, French, and Jones (2010) showed that medical expense risk is a principal determinant of precautionary saving, particularly among higher-income elderly singles. Third, the consumer can trade in a complete annuity market spanning the full space of state-contingent claims (Davidoff, Brown, and Diamond, 2005). Fourth, Yaari's single-good utility specification assumes state-independent preferences – the marginal utility of consumption does not depend on health status – contradicted by evidence of state-dependent utility, though the direction is contested: Finkelstein, Luttmer, and Notowidigdo (2013) find marginal utility declines with health deterioration, though Lillard and Weiss (1997) find the opposite and Brown, Goda, and

McGarry (2016) find heterogeneous effects across health conditions. Market completeness is not an assumption of Yaari's original model but a sufficient condition introduced by Davidoff, Brown, and Diamond (2005) under which the full-annuitization result generalizes beyond Yaari's specific preference structure.

Table 1 enumerates the nine assumptions whose violation reduces or eliminates predicted annuitization.

Table 1. Nine Assumptions of the Yaari Benchmark Whose Violation Reduces Predicted Annuitization

No.	Assumption	Status in Yaari (1965)	Key Paper(s) Identifying Importance	Empirical Status
1	Expected utility (von Neumann-Morgenstern)	Explicit	Hu and Scott (2007); Brown et al. (2008)	Strong evidence against: systematic violations including loss aversion, probability distortion, and framing dependence
2	Additively separable utility	Explicit	Bommier and Le Grand (2014)	Strong evidence against: evidence of habit formation and intertemporal nonseparability
3	No bequest motive	Explicit	Lockwood (2012); Kopczuk and Lupton (2007)	Strong evidence against: approximately 75% of elderly singles estimated to have bequest motives (Kopczuk and Lupton, 2007); 67% of those with children report leaving inheritance as important (Lockwood, 2012)
4	Actuarially fair pricing	Explicit	Mitchell et al. (1999)	Not satisfied for population mortality: MWR 0.80-0.85; closer on annuitant mortality: MWR 0.90-0.94
5	Known, fixed survival function	Explicit	Reichling and Smetters (2015); O'Dea and Sturrock (2023)	Strong evidence against: mortality is stochastic and correlated with health expenditures; individuals systematically underestimate survival probabilities

No.	Assumption	Status in Yaari (1965)	Key Paper(s) Identifying Importance	Empirical Status
6	Exponential discounting	Standard in applications (Yaari uses a general discount function)	Laibson (1997)	Contested: evidence of present bias and quasi-hyperbolic discounting
7	No pre-existing annuitized income	Implicit	Dushi and Webb (2004); Bernheim (1991)	Strong evidence against: Social Security and pensions constitute 50-75% of retiree wealth
8	No expenditure shocks beyond mortality	Implicit	De Nardi, French, and Jones (2010); Reichling and Smetters (2015)	Strong evidence against: stochastic medical costs (combined out-of-pocket and Medicaid) with mean rising from $5,100 at age 70 to $29,700 at age 100 and 95th percentile reaching $111,200/year at age 100 (Jones et al., 2018; in 2014 dollars)
9	State-independent utility	Implicit (in the single-good consumption specification)	Finkelstein, Luttmer, and Notowidigdo (2013); Brown, Goda, and McGarry (2016)	Contested in direction: strong evidence of state dependence, but sign varies – FLN (2013) find marginal utility declines with health deterioration; Lillard and Weiss (1997) find the opposite; Brown et al. (2016) document heterogeneity across health conditions

Note: The "Empirical Status" column summarizes evidence on the descriptive realism of each modeling assumption; entries reflect the weight of available evidence rather than formal hypothesis tests. Assumptions 1-5 are explicit conditions of Yaari's proof, with 1 and 2 relaxable under complete markets (Davidoff, Brown, and Diamond, 2005). Assumption 6 (exponential discounting) is standard in the applied literature but not an explicit condition of Yaari's general formulation. Assumptions 7-9 are features of Yaari's modeling environment –

initial endowments, state space, and utility specification – that subsequent research identified as empirically load-bearing determinants of the full-annuitization prediction. All nine are maintained in Yaari's setup and each, when replaced with empirically grounded alternatives, reduces predicted voluntary annuitization.

Each assumption represents a dimension along which the full-annuitization result can break down. Davidoff, Brown, and Diamond (2005) showed that some of Yaari's preference assumptions (1, 2, and 6) can be relaxed if complete markets are assumed, but as Section 2.3 discusses, this shifts the load-bearing assumption rather than eliminating it. For any real-world retiree – who holds bequest motives, receives Social Security, faces health expenditure risk, and evaluates decisions with reference-dependent preferences – multiple assumptions are violated simultaneously, and the theoretical case for full annuitization weakens accordingly.

2.3 The Davidoff, Brown, and Diamond (2005) Generalization

Davidoff, Brown, and Diamond (2005) showed that with no bequest motive, complete markets, and a return premium net of costs, full annuitization is optimal regardless of utility function – under complete markets, any consumption plan achievable without annuities can be replicated with annuities at lower cost.

This is a significant result: DBD established that full annuitization is robust to preference structure, conditional on complete markets. The preference assumptions relaxed represent a genuine generalization. However, the assumption retained and elevated – complete markets – is among the most counterfactual in economics. Complete markets require that consumers can trade in state-contingent claims spanning every relevant source of uncertainty: longevity, health expenditure, inflation, long-term care costs, and more. Real annuity markets offer a narrow menu of products, predominantly fixed nominal annuities, spanning only the longevity dimension. The generalization shifted the load-bearing assumption from several modestly unrealistic conditions to a single, dramatically unrealistic one. When markets are incomplete, the budget-set dominance argument fails because some consumption plans are achievable only through non-annuitized wealth. Davidoff, Brown, and Diamond also established a weaker result – that some positive level of annuitization is welfare-improving under broader conditions – consistent with this article's position that annuities are not valueless, but that the prediction

of full annuitization is an artifact of extreme assumptions. DBD's incomplete-markets result is important: it implies that the empirical question is not whether any annuitization is optimal (it usually is), but whether full annuitization is descriptively realistic as a benchmark prediction.

2.4 The Knife-Edge Characterization

Perturbation along any of the nine dimensions reduces optimal annuitization; along several, calibrated models predict demand at or near zero. Bequest motives drive willingness to pay for loaded annuities to zero (Lockwood, 2012). Stochastic mortality correlated with medical costs drives optimal demand to zero or negative, even without bequests and under fair pricing (Reichling and Smetters, 2015). Under cumulative prospect theory with standard parameters and narrow framing, predicted demand falls to zero, though the result is sensitive to reference point specification and the loss aversion coefficient (Hu and Scott, 2007). Replacing separable utility with the multiplicative specification of Bommier and Le Grand (2014), for sufficiently high lifetime risk aversion combined with any positive bequest motive, inverts the standard comparative static: greater risk aversion produces lower annuity demand.

These perturbations operate independently; in reality, they operate simultaneously. A typical retiree holds bequest motives, receives Social Security, faces stochastic health expenditure risk, evaluates decisions with reference-dependent preferences, and confronts products priced 15-20 percent above actuarial fair value. The set of parameter configurations at which full annuitization remains optimal is narrow relative to the space of empirically plausible calibrations – most calibrated models reviewed in Sections 3-6 produce demand well below the Yaari benchmark, though Peijnenburg et al. (2016) demonstrate that full annuitization can survive some relaxations under specific parameterizations. The full-annuitization result is knife-edge in the following specific sense: not that infinitesimal perturbations destroy partial-annuitization results – Davidoff, Brown, and Diamond (2005) showed that some positive annuitization remains optimal under broader conditions – but that the conjunction of assumptions required for *full* annuitization occupies a narrow region of the empirically plausible parameter space. This characterization is central to the survey's argument: the annuity "puzzle" is the gap between a fragile theoretical prediction and robust empirical

reality. When the prediction rests on nine restrictive assumptions, none of which holds unchallenged as a realistic description of retirement decision-making, a reasonable inference is that the model is inadequate as a descriptive benchmark, not that consumer behavior is anomalous. The remaining sections demonstrate this proposition channel by channel, with the interaction effects analyzed in Section 7 suggesting that the channels compound rather than merely add.

3. Demand-Side Resolution I: Pre-Existing Annuitization and Bequest Motives

Once the analysis incorporates the annuitized wealth that households already possess through Social Security and defined-benefit pensions, and once it accounts for the widespread preference to leave wealth to descendants, predicted voluntary private annuity demand converges toward observed levels. These two channels alone substantially narrow the gap between theoretical prediction and observed behavior.

3.1 The pre-annuitization baseline

The standard methodology for documenting the annuity puzzle compares private annuity wealth to financial wealth, understating the degree to which retirees are already insured against longevity risk. The relevant object is the share of total retirement resources already in life-contingent form – including Social Security and defined-benefit pensions, which are lifelong and largely non-commutable (Social Security is inflation-indexed by statute and non-commutable; DB pensions vary, with many providing limited or no cost-of-living adjustment, and some plans offering lump-sum distribution options). To measure "annuitization" while excluding them is analogous to measuring housing consumption while ignoring the house in which the household lives.

The quantitative significance is large. Lockwood (2012) reports that households in the bottom quartile have approximately 75 percent of total wealth already annuitized; at the median, approximately 67 percent; even at the 80th-90th percentiles, roughly half. Only in the top decile does the share fall to approximately 33 percent. Private annuity payments

account for roughly one percent of total household income among Americans aged 65 and older (Johnson, Burman, and Kobes, 2004). Dushi and Webb (2004) confirmed that "much of the failure of the average currently retired household to annuitize can be attributed to the exceptionally high proportions of the wealth of these cohorts that is pre-annuitized." Lockwood (2012) concurs: "people in the lower half of the wealth distribution gain little from annuities even without bequest motives because they are fairly well annuitized by Social Security and employer pensions." Bernheim (1991) found that increases in Social Security benefits raised life insurance holdings while depressing private annuity demand, and Pashchenko (2013) confirms in a structural model that once Social Security is accounted for, optimal additional private annuitization drops substantially.

Because Social Security benefits are progressive and lower-wealth households hold a larger share in Social Security form, the annuitized share tends to be inversely related to financial wealth. This compositional gradient is one contributor among several to the wealth pattern in annuity demand; differential mortality by socioeconomic status (higher-income groups live longer, partially offsetting formula progressivity), variation in claiming behavior, and differences in DB pension coverage also shape the distribution. Poterba, Venti, and Wise (2011) find that Social Security and defined-benefit wealth constitute approximately 46 percent of mean household wealth at retirement, though this average masks substantial heterogeneity: approximately 30 percent of retirement-age households hold more than $518,000 in non-annuitized wealth. Johnson, Burman, and Kobes (2004) estimate that public and private annuities combined represent approximately 41 percent of total household wealth for elderly adults. An important caveat: these estimates reflect cohorts with substantial DB pension coverage. As private-sector DB coverage declines – currently approximately 14 percent of private-industry workers in the United States have access to a defined-benefit plan (Bureau of Labor Statistics, 2025, Table 1) – the pre-annuitization share for future retirees will be lower, resting principally on Social Security alone, and the dissolution thesis becomes correspondingly more cohort-dependent (Section 8.4). The residual question concerns the top 10-20 percent: why do the wealthiest retirees not annuitize more? As the next subsection demonstrates, bequest motives provide a compelling answer.

3.2 Bequest motives

Yaari's (1965) theorem assumes the consumer places zero value on wealth remaining at death. Relaxing this single assumption proves sufficient to align theoretical predictions with observed behavior, and the empirical prevalence of bequest motives is well established. Kopczuk and Lupton (2007) estimate, using structural estimation of a life-cycle consumption model for elderly singles, that approximately 75 percent have a bequest motive, and that roughly half of net wealth transferred at death reflects intentional preferences rather than accidental residuals. Among those with children, 67 percent report that leaving an inheritance is important; even among those without children, 55 percent express a desire to leave bequests (Lockwood, 2012).

The quantitative impact on annuity demand is large but requires disaggregation across two margins. Lockwood (2012) estimates a lifecycle model calibrated to U.S. data. Without bequest motives, the model predicts 61 percent ownership among single retirees aged 65-69, with willingness to pay (WTP) of 25.3 percent of non-annuity wealth for actuarially fair contracts. With moderate bequest motives – calibrated to imply bequests of roughly one-fifth of wealth – WTP for fair annuities collapses to 3.7 percent, and WTP for annuities carrying a 10 percent load falls to zero. The empirical ownership rate among single retirees aged 65 to 69 is approximately 3.6 percent.

Two distinct objects are at work. On the intensive margin (WTP for annuity access), bequest motives alone produce a large reduction robust across specifications. On the extensive margin (predicted ownership rates), the match to observed data depends not only on bequest parameterization but also on additional market frictions. In Lockwood's simulations, moderate bequest motives without a fixed purchase cost typically reduce predicted ownership to approximately 10-15 percent; it is the addition of a fixed purchase cost (approximately $2,000) that pulls predicted ownership to the 4-5 percent range matching the observed 3.6 percent. With the stronger bequest parameterization of Ameriks, Caplin, Laufer, and Van Nieuwerburgh (2011), predicted ownership falls to zero. The directional effect – that bequest motives dramatically reduce predicted annuitization – is robust across specifications, though the precise quantitative match depends on the specific bequest parameterization employed.

The mechanism is revealing. Lockwood's (2012) decomposition shows that in the no-bequest model, more than 75 percent of the welfare gains from annuitization derive from reclassifying accidental bequests as "waste." When the consumer places no value on wealth at death, every dollar left behind represents foregone consumption, and annuities convert this waste into spending. But once bequests enter the utility function, wealth left to heirs is a valued outcome. The mortality credit, which appears as free money in the no-bequest framework, is more accurately described as the price of forfeiting a valued bequest. With even moderate preferences, this price exceeds the benefit.

The interaction between bequest motives and annuity pricing loads further strengthens the result. Davidoff, Brown, and Diamond (2005) showed that under actuarially fair pricing, even consumers with moderate bequest motives should annuitize some wealth. But private annuities carry loads of approximately 15-20 percent against population mortality tables (Mitchell et al., 1999). Lockwood (2012) demonstrates that bequest motives and realistic loads are multiplicative rather than additive in suppressing demand: consumers whose bequest preferences alone would leave them indifferent to fair annuities are driven decisively away once the load is introduced. Kotlikoff and Spivak (1981) show that married couples can achieve approximately 70 percent of the welfare gains from fair annuity markets through intra-household risk sharing, and Brown and Poterba (2000) confirm that utility gains from annuitization are substantially lower for married couples than for singles – a significant qualification given that the majority of wealth at retirement is held by married households. Tax asymmetries further reinforce the bequest channel: in the United States, inherited financial assets receive a step-up in cost basis, eliminating unrealized capital gains tax, whereas annuity payouts are taxed as ordinary income. This creates an implicit tax penalty for annuitization that compounds the bequest motive's demand-suppressing effect. Finally, housing wealth – the largest un-annuitized asset for the median retiree – serves as an endogenous substitute for formal annuitization, functioning simultaneously as a precautionary buffer (accessible via downsizing or reverse mortgage) and a bequest vehicle, thereby reducing the marginal value of converting financial wealth into annuity income.

3.3 Evolutionary perspectives on preferences

Evolutionary psychology provides ultimate explanations – rooted in fitness consequences – for the preference parameters that structural and behavioral economists treat as proximate inputs, such as the 75 percent bequest-motive prevalence estimated by Kopczuk and Lupton (2007). Hamilton's (1964) theory of inclusive fitness predicts that natural selection favors downward resource transfers to genetic relatives – a prediction confirmed by human life-history research documenting extensive intergenerational transfers across economic systems (Hooper, Gurven, Winking, and Kaplan, 2015; Borgerhoff Mulder et al., 2009; Kaplan et al., 2000). The specific form of at-death financial bequest is culturally contingent, but the underlying motivational substrate is widespread: zero-bequest assumptions are not merely empirically false but anthropologically implausible as a description of a species shaped by kin selection (Hamilton, 1964; Hooper et al., 2015).

The evolutionary framework also explains why the behavioral departures documented in Section 5 – loss aversion, narrow framing, present bias, survival pessimism – exist and persist. These are not random cognitive defects but evolved heuristics that were adaptive in ancestral environments of high mortality and immediate threats, and that generate predictable errors in modern financial contexts where the relevant tradeoffs (irrevocable commitment, 20-year time horizons, probabilistic longevity) have no ancestral analogue (Cosmides and Tooby, 1994; Friedman, 2004). This carries a practical implication: if the behavioral biases suppressing annuity demand reflect deeply evolved decision architecture rather than correctable ignorance, then financial literacy interventions are less likely to increase demand than institutional design – defaults, product simplification, and choice architecture – consistent with the evidence from the UK pension freedoms and TIAA-CREF cases discussed in Section 6.5.

4. Demand-Side Resolution II: Health Expenditure Risk and Stochastic Mortality

This section identifies a third channel: the risk of catastrophic health and long-term care expenditures. Annuities convert liquid wealth into an illiquid income stream, yet retirees face medical costs that are large, unpredictable, and concentrated in states where liquidity is most

valuable.

4.1 The liquidity imperative

Health expenditure risk in retirement is large and concentrated in the right tail. Jones, De Nardi, French, McGee, and Kirschner (2018) estimate that mean annual out-of-pocket medical expenditures among surviving households (in 2014 dollars) average approximately $4,200 at age 70; combined out-of-pocket and Medicaid expenditures rise from approximately $5,100 at age 70 to $29,700 by age 100, with the 95th percentile of combined annual expenditures reaching approximately $111,200 at age 100. The liquidity threat to asset-holding households – the primary candidates for voluntary annuity purchase – is concentrated in out-of-pocket costs, since Medicaid payments accrue only after asset spend-down, a path that most potential annuity purchasers seek to avoid. The distribution is heavily right-skewed: most retirees face manageable costs, but a substantial minority confront expenditures that would exhaust all but the largest portfolios.

An important institutional distinction bears on this analysis. Medicare covers the majority of acute care expenditures for Americans aged 65 and older; the dominant uninsured tail risk is long-term care – nursing home stays, assisted living, and home health care – which Medicare covers only for short post-acute episodes. The median annual cost of a semi-private nursing facility room was $111,325 in 2024 (Genworth, 2024), and approximately 9 percent of those in the 2020 cohort turning 65 were projected to spend five or more years in a nursing home; for the 2000 cohort, where gender-specific estimates are available, the figures were 4 percent of men and 12 percent of women (Spillman and Lubitz, 2002, projections based on then-current utilization patterns specific to nursing-home stays rather than all forms of long-term care). These costs are lumpy, unpredictable, and positively correlated with deteriorating health – precisely the profile for which liquid wealth is most valuable and annuitized income least adequate.

4.2 Precautionary saving and public care aversion

De Nardi, French, and Jones (2010) show that medical expense risk drives substantial precautionary wealth accumulation, particularly among higher-income elderly singles – explaining

the slow decumulation that the standard lifecycle model without health risk cannot account for. Ameriks, Caplin, Laufer, and Van Nieuwerburgh (2011), using strategic survey methods on a web-based sample (n = 938 after screening; structural estimation uses the 498 single respondents, following De Nardi, French, and Jones, 2010) to separate bequest motives from "public care aversion" – the desire to avoid reliance on Medicaid-funded care – find that both motives are economically significant. Public care aversion is widespread and particularly helps explain low spending among middle-class retirees, while bequest motives remain quantitatively important across the wealth distribution. Retirees retain liquid wealth not because they fail to understand longevity insurance, but because they rationally prefer to maintain the capacity to pay for high-quality care.

The Medicaid interaction reinforces this behavior through a mechanism related to, but not identical to, the implicit tax Brown and Finkelstein (2008) documented for long-term care insurance. Brown and Finkelstein estimate that Medicaid imposes an implicit tax of 60-75 percent on private LTC insurance for median-wealth individuals, because private benefits paid in the Medicaid-eligible state partially substitute for public payments. The mapping to annuity demand differs because annuity payouts are not contingent on needing care; nevertheless, means-tested public LTC coverage affects the value of annuitization through both eligibility and post-eligibility channels. First, annuitizing converts liquid wealth into an irrevocable payment stream, reducing the option value of holding spendable assets to self-fund high-quality care and thereby avoid (or delay) Medicaid reliance. Second, conditional on entering Medicaid-financed nursing home care, most income – including annuity income, net of small allowances – is typically applied to the cost of care, so the marginal value of annuity payouts in the high-expense state can be low because they effectively reduce Medicaid's payment obligation. These effects are heterogeneous, depending on household structure (single versus married under community spouse rules), annuity contract features, and state-specific Medicaid treatment of annuities and income. While some households can use Medicaid-compliant immediate annuities as part of crisis planning, the broader implication for the annuity puzzle literature is that means-tested public LTC coverage lowers the marginal value of annuitized income in precisely the states where spending flexibility is most valuable, reinforcing the public care aversion that Ameriks et al. (2011) identify as a primary driver of

precautionary wealth retention.

4.3 Stochastic mortality and the Reichling-Smetters result

Reichling and Smetters (2015) provide the most powerful formulation. They allow health shocks to alter survival probabilities stochastically: a negative health shock simultaneously increases the marginal utility of wealth (through elevated medical costs) and reduces the remaining present value of annuity payments (through shortened life expectancy). The key mechanism is the correlation structure: the annuity becomes "valuation-risky" because a health shock generates implicit depreciation in annuitized wealth exactly when liquid resources are most valuable.

This correlation-driven mechanism – not generic "medical expense risk" – undermines the core value proposition of annuitization. Reichling and Smetters explicitly note that lumpy expenditures uncorrelated with health have no impact on optimal annuitization in their framework; indeed, in some specifications generic medical expense risk can even increase annuity demand (Pashchenko, 2013), because longevity insurance covers old-age states when health costs are highest; the demand-suppressing result depends specifically on the joint distribution of health shocks, survival probabilities, and medical costs. The mortality credit is largest for individuals who live longest, but those who live longest are, on average, those who remain healthy and face lower medical costs. When this health-mortality-cost correlation is calibrated to realistic magnitudes, optimal annuity demand falls to zero or even becomes negative – consumers may optimally wish to hold negative annuity positions (via life insurance) to hedge the joint risk of health deterioration and shortened lifespan.

This result is particularly significant because it operates even under actuarially fair pricing and even absent bequest motives – conditions under which the Yaari result and its Davidoff, Brown, and Diamond (2005) generalization both predict full annuitization. The case for annuitization depends not only on the existence of longevity risk, which annuities address, but on the absence of correlated health-expenditure risk, which annuities exacerbate. The Reichling-Smetters result is thus not merely another friction reducing demand at the margin; it identifies a structural channel through which the core insurance value proposition of annuitization is undermined.

Peijnenburg, Nijman, and Werker (2016), examining multiple frictions jointly in a calibrated lifecycle model, find that full annuitization remains approximately optimal even after incorporating incomplete markets, background risk, bequest motives, and default risk – leading the authors to title their paper "The Annuity Puzzle Remains a Puzzle." The divergence with this survey is methodological: Peijnenburg et al. evaluate channels within a single unified model, while the present survey synthesizes across modeling frameworks. The Peijnenburg et al. finding that no single channel suffices reinforces the emphasis on compounding interaction effects analyzed in Section 7.

5. Behavioral Models: From Expected Utility to Prospect Theory

The preceding sections relaxed the Yaari framework's assumptions within the expected utility paradigm. This section asks what happens when the behavioral model itself is replaced. Under standard behavioral parameterizations, predicted annuity demand can fall to zero.

5.1 Framing Effects and Description Invariance

Brown, Kling, Mullainathan, and Wrobel (2008) demonstrated that description invariance – a foundational EU axiom – fails sharply in the annuity context. Approximately 72 percent of respondents preferred the annuity when framed as consumption insurance, versus only 21 percent when framed as an investment. The 51-percentage-point swing is strong evidence against description invariance in stated annuitization preferences. Brown et al. employed hypothetical choices, and external validity for a decision this consequential warrants caution. Nevertheless, the directional finding – that framing materially influences stated annuitization preferences – has been replicated across subsequent designs (Agnew et al., 2008; Beshears et al., 2014), though effect sizes vary and not all frame manipulations produce significant effects.

The theoretical significance is substantial. If preferences shift by a factor of more than three depending on how the choice is represented, then low observed demand is difficult to reconcile with stable underlying preferences of the kind expected utility theory presupposes. Demand is, to a considerable degree, constructed by the frame – and in many real-world settings, the frame is the investment frame. Retirees face accumulated lump sums in defined-

contribution accounts and are asked whether to surrender them for an income stream. Industry materials emphasize internal rates of return and breakeven ages, coding the annuity as a bet rather than as insurance.

5.2 Prospect Theory and Predicted Annuity Demand

The framing results find a natural theoretical home in prospect theory. Kahneman and Tversky (1979) proposed that individuals evaluate risky prospects not in terms of final wealth levels but as gains and losses relative to a reference point. The value function is concave over gains and convex over losses, steeper for losses than for equivalent gains (loss aversion), and applied not to objective probabilities but to decision weights generated by a nonlinear probability weighting function. In its cumulative form (Tversky and Kahneman, 1992), the model is parameterized by a loss aversion coefficient of approximately 2.25, a value function exponent of 0.88, and probability weighting functions that overweight small probabilities and underweight moderate-to-large ones.

Hu and Scott (2007) were the first to apply this framework formally to the annuity decision. Their central analytical object is a reservation price – the maximum fraction of actuarially fair cost that a CPT agent would be willing to pay for an annuity. Reservation prices below 1 imply that the annuity is unattractive even at actuarially fair pricing. Under the full CPT specification with standard parameters and narrow framing (evaluating the annuity as a standalone gamble), reservation prices for a 65-year-old male fall to approximately 0.77, implying zero demand at fair or loaded prices. Three mechanisms reinforce: loss aversion weights early-death losses at 2.25 times equivalent longevity gains; probability weighting overweights the small probability of dying soon after purchase while underweighting the moderate probability of surviving into the late eighties; and diminishing sensitivity compresses gains from extreme longevity.

Three sources of sensitivity condition this result. First, the loss aversion coefficient varies substantially across studies, with estimates typically ranging from 1.5 to 2.5 though specification-dependent estimates can be considerably higher, and probability weighting curvature also varies. Second, the reference point specification is the most consequential modeling choice: Hu and Scott assume narrow framing (evaluating the annuity as a standalone

gamble); under broader integration with background wealth, the zero result need not hold. Third, with no time discounting, reservation prices can exceed 1, making annuities attractive even for CPT agents. The zero-demand result is most robust under narrow framing, moderate-to-high discounting, and standard parameters simultaneously. An additional caveat is that CPT was developed for single-period gambles; Hu and Scott's application extends it to a multiperiod consumption stream, and as they note, their results should be interpreted qualitatively given the cognitive demands of intertemporal CPT evaluation.

The qualitative finding – that prospect theory substantially reduces predicted annuity demand relative to expected utility – holds across a wide range of plausible parameter values, even if the specific zero result depends on the baseline parameterization and framing assumptions. What Hu and Scott's analysis most clearly establishes is that under narrow framing and standard experimental parameters, CPT agents find actuarially fair annuities unattractive – a result whose field relevance depends on whether the narrow-framing assumption accurately describes how retirees evaluate annuity decisions in practice. An important product-specific nuance: Hu and Scott's own Table 1 shows that far-deferred longevity annuities can be attractive even under full CPT – a 65-year-old's reservation price for an annuity beginning payouts at age 95 reaches 1.20, because probability distortion overweights the small survival probability. The directional finding – that CPT's zero-demand prediction is specific to immediate annuities and does not necessarily extend to deferred products – is relevant to the deferred income annuity puzzle discussed in Section 8.1.

Brown, Kapteyn, Luttmer, and Mitchell (2017) provided complementary evidence using a large-sample survey administered through the American Life Panel. They found that the prices at which individuals are willing to buy annuities are substantially below those at which they are willing to sell them – a buy-sell spread inconsistent with coherent expected-utility valuation. Under EU, the maximum willingness to pay and minimum willingness to accept should converge; the persistent gap implies that the annuity is simultaneously valued below its cost as a purchase (coded as a loss of liquid wealth) and above its cost as an existing endowment (coded as a loss of income if surrendered). This buy-sell asymmetry is consistent with the reference-dependence mechanism that Hu and Scott (2007) modeled theoretically, though the authors emphasize cognitive constraints on annuity valuation as the primary

interpretation.

5.3 Prospect Theory Explains Observed Market Patterns

Gottlieb (2012, working paper; not yet published in a peer-reviewed journal) argues that a single CPT framework can account for three features that EU models find difficult to generate simultaneously: the prevalence of guarantee and period-certain features that cap downside loss from early death; simultaneous holding of life insurance and annuities (under narrow framing, each is evaluated as a separate gamble satisfying loss-limiting preferences); and persistent consumer preference for loss-limiting riders over pure life annuities offering higher payments. Prospect theory offers a parsimonious descriptive account of these patterns, though EU alternatives (bequests, lifecycle needs, incomplete markets) can explain some individually.

5.4 Survival Pessimism

O'Dea and Sturrock (2023) documented that individuals systematically underestimate their own survival probabilities – by approximately 25 to 30 percentage points among those in their early sixties estimating survival to age 75, and by 15 to 20 percentage points among those in their late sixties and early seventies estimating survival to age 85. Using a lifecycle model calibrated to English Longitudinal Study of Ageing (ELSA) data, they show that subjective survival pessimism markedly reduces perceived annuity value: when individuals believe they are unlikely to survive to advanced ages, the expected present value of annuity payments – as perceived by the decision-maker – falls well below the premium, making non-purchase appear rational from the individual's perspective. This channel operates within the EU framework with distorted inputs rather than distorted preferences, and is analytically distinct from the CPT mechanisms in Section 5.2. Groneck, Ludwig, and Zimper (2016) reach a related conclusion in the consumption-saving domain: using a lifecycle model with ambiguous survival beliefs and non-additive probability measures, they show that pessimism about longevity substantially reduces wealth accumulation. While Groneck et al. do not directly model annuity demand, their framework implies that ambiguity about survival would also reduce the perceived value of longevity insurance.

5.5 Bounded Rationality and Complexity

The annuity purchase is among the most computationally demanding decisions a household faces: it requires estimating one's remaining lifespan distribution, understanding discount rates and the time value of money, evaluating actuarial fairness, accounting for inflation erosion over decades, integrating the annuity with existing income sources, comparing across product structures and providers, and assessing insurer solvency. Each task individually exceeds the demonstrated financial capabilities of the median retiree (Lusardi and Mitchell, 2011); collectively, they describe an optimization problem that would challenge a professional actuary. Complexity interacts with status-quo bias to compound demand suppression (Samuelson and Zeckhauser, 1988). In most institutional settings, the status quo is to retain the lump sum, and there is no institutional nudge toward annuitization comparable to the automatic enrollment that has transformed 401(k) participation rates (Madrian and Shea, 2001).

5.6 The Descriptive-Normative Distinction

The behavioral findings carry an important caveat. Prospect theory, framing effects, and bounded rationality are descriptive models, not normative endorsements. Loss aversion is a psychological regularity, not – by itself – a settled welfare criterion, though Kőszegi and Rabin (2006) and others have incorporated reference-dependent preferences into welfare analysis. As Bernheim and Rangel (2009) argued, frame-dependent preferences cannot straightforwardly serve as the basis for welfare analysis. Chetty (2015) offers a complementary pragmatic approach: using behavioral evidence to bound welfare-relevant parameters without requiring full commitment to a contested structural model, a methodology well-suited to the heterogeneous welfare question identified in Section 8.2.

This paper's normative claims therefore rest on the structural channels documented in Sections 3 and 4. The welfare criterion implicitly employed is expected utility with empirically calibrated parameters – the framework of Lockwood (2012) and Reichling and Smetters (2015) – under which realistic bequest motives, pre-existing annuitization, health expenditure risk, and unfair pricing independently justify low annuitization for a rational agent. For classic immediate life annuities, the behavioral evidence plays a confirmatory rather than

load-bearing role; for deferred income annuities, where structural channels are substantially attenuated, behavioral factors must bear more of the explanatory weight (Section 8.1). CPT-predicted choices should accordingly be described as "predicted demand under CPT" rather than "optimal annuitization."

The convergence of structural and behavioral predictions – both classes of models predict low voluntary annuitization – strengthens the positive case that the aggregate annuity puzzle is dissolved. However, convergence in predicted choices does not establish convergence in welfare implications. Two points of caution are essential. First, if rational low take-up is driven by high loadings, inflation risk, and illiquidity, the welfare implication is "current products are inadequate" – a market design failure that could in principle be remedied. If behavioral low take-up is driven by loss aversion or narrow framing, the welfare implication may instead be that better framing, defaults, or simplified products could improve outcomes. These mechanisms can coexist, and observing that both push toward low annuitization does not identify which dominates welfare. Second, in a second-best world, low annuitization can be simultaneously individually rational given the existing product menu and evidence of large welfare losses from market design failures – the absence of adequately priced real annuities, the lack of annuity-LTC bundles, or regulation-induced pricing distortions. Low annuitization may not be a "consumer mistake" and yet remain a welfare problem in the form of unaddressed insurance-market incompleteness.

The convergence argument is most compelling for households with significant pre-existing annuitization, operative bequest motives, or meaningful health expenditure risk – characteristics that describe the majority of retirees. For the subset lacking these characteristics (single, childless, healthy, wealthy, with low bequest motives), the structural channels provide weaker justification for non-annuitization. Whether behavioral factors account for welfare-reducing under-annuitization for such households depends on the pricing environment: at retail loads of approximately 0.80-0.85 on a population-mortality basis (Mitchell et al., 1999; Wettstein et al., 2021, confirming stability over 2001-2019), even these households may rationally decline; at group or institutional pricing (MWR 0.90+), behavioral barriers could generate meaningful welfare losses. Whether this subpopulation is large enough to constitute a policy-relevant concern is an empirical question addressed further in Section 8.2. Aggregate take-up approxi-

mately matching a model's prediction does not establish allocative efficiency at the individual level: some households may be biased toward annuitization while others are biased away, with offsetting errors producing an aggregate that appears "about right." The distinction between the aggregate positive finding and the heterogeneous normative question is central to this paper's argument.

6. Supply-Side Failures and Cross-Cultural Evidence

This section turns to the supply side. The annuity in Yaari's model – actuarially fair, inflation-indexed, default-free, and frictionless – bears little resemblance to the annuity available in real markets.

6.1 Pricing and Money's Worth

The foundational empirical study of annuity pricing remains Mitchell, Poterba, Warshawsky, and Brown (1999). Using actual quotes from the US individual annuity market and discounting projected payment streams at Treasury yields, they report money's-worth ratios (MWRs) – the expected present value of payouts per premium dollar – of roughly 0.80-0.85 against population mortality tables and roughly 0.90-0.94 against annuitant mortality tables. Mitchell et al. (1999, p. 1310) report that for a 65-year-old male, the gap between population-table and annuitant-table MWRs is approximately 11 cents – "roughly half" of the total shortfall – reflecting the longer lives of voluntary annuity purchasers relative to the general population. The residual shortfall of the annuitant-table MWR from unity reflects insurer expenses, statutory reserves, distribution costs, taxes, and profit. Both components vary meaningfully by age, sex, and assumed discount rate. Wettstein, Munnell, Hou, and Gok (2021) found this overall pricing wedge remarkably stable over the period 2001-2019, with levels comparable to late-1990s estimates from Mitchell et al., with the population-table MWR fluctuating between 0.76 and 0.87 for men during 2001-2019. They also documented a growing disparity across socioeconomic groups: by 2019, the MWR gap between top- and bottom-education-tercile white men had grown to 14 cents, implying that less-educated, lower-income buyers face substantially lower money's worth under uniform pricing that does not reflect their shorter

average lifespans.

The MWR picture is sensitive to the interest rate environment. The 2022-2024 rate cycle meaningfully improved SPIA payout rates. Total annuity industry sales reached record levels, driven primarily by fixed and fixed-indexed products (LIMRA, 2025). However, voluntary SPIA and longevity annuity purchases did not increase commensurately – reinforcing the thesis that demand suppression reflects structural preferences rather than pricing alone.

Adverse selection compounds the pricing problem. Finkelstein and Poterba (2002), using the UK's dual voluntary-compulsory market structure, demonstrated that adverse selection is roughly twice as severe in voluntary markets as in compulsory markets. Finkelstein and Poterba (2004) showed that selection operates along contract structure dimensions – particularly back-loading and estate-payment features – but found little to no selection along payment size, the variable that standard adverse selection models emphasize. The authors caution that "it is unlikely that this adverse selection, and the associated high effective prices for annuities, can fully explain the limited demand for voluntary annuities" (Finkelstein and Poterba, 2004, p. 186). The consequence is a self-reinforcing equilibrium: prices reflect a pool dominated by long-lived individuals, deterring average-mortality buyers and further concentrating the pool.

An important distinction: the MWR is a pricing metric, not a welfare measure. MWR $<$ 1 does not automatically imply welfare-reducing purchase – for sufficiently risk-averse agents, the insurance value of the mortality credit can dominate the pricing shortfall. But additional frictions – inflation erosion (Section 6.2), counterparty risk, illiquidity, and product complexity – further reduce the welfare-equivalent value below the nominal MWR. For consumers with calibrated bequest motives, health risk, or behavioral frictions, non-annuitization is individually rational even under expected utility maximization.

6.2 Inflation Risk

Yaari's theorem implicitly assumes a real annuity. Most annuities sold are nominal: at 3 percent inflation, a nominal annuity loses roughly 45 percent of purchasing power by year 20, delivering its weakest protection in late old age when medical costs are highest. Inflation-indexed annuities are rare, and initial payouts are typically approximately 30 percent

lower than comparable nominal products (30 percent for men, 33 percent for women; Brown, Mitchell, and Poterba, 2002) – though this largely reflects backloading of real purchasing power rather than additional expense, with the welfare cost of restricting retirees to nominal rather than real annuities being relatively modest in utility terms (Brown, Mitchell, and Poterba, 2001, 2002). Nevertheless, the lower initial payout may trigger loss aversion relative to the higher nominal alternative, and the limited supply of long-duration inflation-linked assets constrains insurer capacity.

6.3 Counterparty Risk and Product Complexity

A fixed annuity constitutes, in economic substance, a long-term, illiquid claim on an insurance company's general account – subject to state insurance regulation, reserve requirements, and guaranty association protections, but without the securities-law investor protections or deposit insurance that apply to other financial instruments. The collapse of Executive Life Insurance Company of California in 1991 left annuitants receiving only 70 percent of monthly payments for nearly 13 months, though subsequent rehabilitation proceedings improved outcomes. Life insurer insolvencies are rare – NOLHGA has coordinated roughly 100 multi-state cases over four decades, with none since the 2008 financial crisis – and average recoveries have exceeded 96 percent on life claims and 94 percent on annuity claims (NOLHGA, n.d.). State guaranty associations provide backstops with coverage typically set at $250,000 in present value of annuity benefits, though caps vary by jurisdiction and some annuitants in past liquidations have experienced shortfalls when contract values exceeded guaranty limits (National Organization of Life and Health Insurance Guaranty Associations [NOLHGA], n.d.).

Product complexity compounds counterparty risk. A definitional clarification is important: the annuity puzzle concerns longevity insurance products (SPIAs and DIAs), but the retail annuity market is overwhelmingly dominated by products serving as tax-advantaged accumulation or principal-protection vehicles. Total U.S. annuity sales reached a record $434.1 billion in 2024 (LIMRA, 2025), but SPIAs accounted for only $13.6 billion and DIAs $4.9 billion – together representing less than 5 percent of total annuity sales. The market's center of gravity has shifted toward fixed-rate deferred annuities ($153.2 billion), fixed indexed

annuities ($126.9 billion), and registered index-linked annuities (RILAs, $65.6 billion). Many consumers are buying annuity products but not purchasing longevity insurance.

Fee structures vary widely across product types, with total annual costs ranging from under 1 percent in fee-based advisory share classes to over 3 percent in traditional variable annuity configurations with optional living-benefit riders. The product complexity literature (Carlin, 2009) establishes that strategic price complexity inhibits comparison shopping, and while technology has reduced search costs for immediate annuities, the complexity of indexed and rider-laden products remains a friction. An important distinction applies: the most prominent fee, commission, and complexity evidence pertains to variable and complex deferred annuities, whereas the annuity puzzle as analyzed in this paper concerns SPIAs and DIAs – simpler products with lower embedded costs and different distribution economics.

6.4 Distribution Channel Conflicts

The incentive structure of the advisory industry creates systematic biases against annuitization. Assets-under-management advisors have a direct financial interest in discouraging annuitization, since every dollar that flows into an annuity leaves the fee-generating portfolio; commission-based advisors face the opposite incentive, favoring complex, high-commission products. The regulatory environment has evolved: the SEC's Regulation Best Interest (effective June 2020) and the NAIC Suitability in Annuity Transactions Model Regulation (revised 2020) have partially eroded the clean AUM-versus-commission dichotomy, and the growth of direct-to-consumer annuity platforms has reduced search costs for some buyers. Nevertheless, conflicted distribution remains a real friction: neither fee-based nor commission-based incentive structures are reliably aligned with client welfare, and the advisory industry's structural biases continue to influence the annuitization decision for the majority of retirees who receive financial advice. The strongest empirical evidence on conflicted distribution concerns variable annuities, where commissions and rider complexity create the most acute misalignment; the evidence base for SPIA and DIA distribution frictions, while directionally consistent, is thinner.

6.5 Cross-Cultural Institutional Variation

If low annuity demand were driven primarily by market imperfections rather than by deeper preferences, we would expect significant variation across institutional contexts. The cross-cultural evidence provides instructive institutional variation. Table 2 summarizes the major cases.

Table 2: Cross-cultural evidence on annuity demand

Context	Policy variation	Measure	Outcome	Source
United Kingdom (2015)	Removed mandatory annuitization ("pension freedoms"); simultaneously introduced flexi-access drawdown	Contract volume (flow)	Quarterly sales fell from 74,092 to 18,243 (~75% decline; ABI data reported in FCA, 2018); annual sales rebounded to 89,600 contracts by calendar year 2024 (ABI, 2024) driven by higher interest rates; confounders include RDR advice market reform and near-zero rate environment	FCA (2018); FCA (2019); ABI (2024)
TIAA-CREF, United States (2000–2018)	Declining institutional norms favoring annuitization	First-time income choice	Fraction of first-time retirement income recipients selecting a life-contingent annuitized payout declined from 61 percent to 18 percent; among those who deferred income until RMD age, the share fell from 41 percent to approximately 6 percent	Brown, Poterba, and Richardson (2025)
Australia (ongoing)	No annuitization mandate in compulsory superannuation system; ABP drawdown as dominant default	Asset stock share	Lifetime annuity products constitute a negligible fraction of approximately AUD 3.6 trillion in superannuation assets; Retirement Income Covenant (effective July 2022) mandates trustees formulate longevity-risk strategies	Australian Treasury (2023)

Context	Policy variation	Measure	Outcome	Source
Singapore (2009/2013)	CPF LIFE introduced voluntarily (2009), mandatory from 2013 for members with sufficient balances	Compliance rate	High compliance under mandatory status; broader CPF policy framework generated public debate	Fong, Mitchell, and Koh (2011); CPF Board
Chile (1981–present)	AFP system with choice between programmed withdrawal and immediate annuity at retirement	Share of new retirees choosing annuity (flow)	Over 60% of eligible retirees purchase private annuities; annuity take-up varies by cohort, wealth, and marital status. Early retirement rules and minimum pension thresholds substantially constrain the choice set, so high take-up partly reflects institutional design rather than purely voluntary choice	Illanes and Padi (2025)
Netherlands (ongoing)	Quasi-mandatory occupational pensions covering ~90% of workforce; limited lump-sum access	Institutional coverage rate	High annuitization by design; occupational pensions (historically DB and collective arrangements) provide lifelong income; system operates collectively rather than through individual retail purchase. The Future of Pensions Act (effective 1 July 2023) mandates transition to contribution-based contracts by 1 January 2028	OECD country profiles

Note: Entries use different metrics (flow counts, transition rates, stock shares, institutional coverage) and are not directly comparable across institutional contexts. The UK entry constitutes the closest approximation to a quasi-experimental design; Chile and the Netherlands illustrate high-annuitization systems with distinct institutional structures; other entries represent cross-sectional observations or institutional time series.

The UK case provides the strongest quasi-experimental evidence. For decades, Britain appeared to be the great exception to the annuity puzzle: retirees annuitized their DC pension savings at rates far higher than elsewhere. What appeared to be robust consumer demand was, in substantial part, an artifact of compulsion: UK tax law effectively required

annuity purchase, with flexible drawdown available only to wealthier individuals. The 2015 pension freedoms constitute the closest approximation to a natural experiment, though several simultaneous changes limit clean causal interpretation. The reforms removed the annuity requirement while introducing flexi-access drawdown with no income floor. The 2013 Retail Distribution Review had restructured the advice market, and the near-zero rate environment (2015-2021) independently suppressed annuity attractiveness. Quarterly sales fell from 74,092 to 18,243, a decline of approximately 75 percent. By 2018/19, only 11 percent of pension plans accessed for the first time were used to purchase an annuity (FCA, 2019). Annual annuity sales rebounded to 89,600 contracts in 2024 (ABI) as rates rose, suggesting pricing partially explains the initial decline's depth. Nevertheless, the directional finding – removing compulsion substantially reduced take-up – is robust.

The TIAA-CREF case tells a parallel story in what should be among the most favorable environments for annuitization: low-cost products with money's-worth ratios well above the retail market average, a trusted institutional brand with decades of reputation, and a participant population of highly educated university and nonprofit employees – precisely the demographic that financial literacy research identifies as best equipped for complex financial decisions. Yet Brown, Poterba, and Richardson (2025) documented that the fraction of first-time retirement income recipients selecting a life-contingent annuitized payout stream declined from 61 percent to 18 percent over the 2000-2018 period. The decline was even more dramatic among those who deferred all income until RMD age, where the annuity share fell from 41 percent to approximately 6 percent. The decline reflects shifting institutional norms: TIAA historically defaulted participants into annuitized payout structures, and as the system moved toward greater participant choice and lump-sum access – including systematic withdrawal arrangements and required minimum distributions – annuitization declined sharply. The TIAA case is particularly informative because it controls for many supply-side frictions – pricing, product complexity, counterparty risk, and distribution channel conflicts – that plague the retail market, isolating the role of preferences, defaults, and behavioral factors.

Australia provides a cross-sectional observation rather than a natural experiment: compulsory superannuation has accumulated approximately AUD 3.6 trillion with no annuitization mandate. Lifetime annuity products exist and are offered by reputable insurers, yet they

constitute a negligible fraction of superannuation assets. The dominant vehicle is the account-based pension (ABP), a regulated drawdown product that preserves flexibility and bequest capacity. The means-tested Age Pension creates complex interactions with annuitization: under post-2019 rules, lifetime annuities receive comparatively favorable treatment, but depending on individual circumstances, purchasing a lifetime annuity can either reduce or increase Age Pension entitlements, complicating any simple "implicit tax" characterization. The Retirement Income Covenant (enacted 2021; effective July 2022), mandating that trustees formulate longevity-risk strategies, implicitly acknowledges that the market has not spontaneously generated adequate longevity products – a fact that complicates the preference-based explanation, though it does not refute it.

Singapore's CPF LIFE, mandatory from 2013, achieved high compliance as a national longevity insurance scheme (Fong, Mitchell, and Koh, 2011). Scandinavian hybrid systems achieve high annuitization through institutional defaults, further illustrating the sensitivity of outcomes to institutional design.

The cross-cultural evidence does not support a uniform narrative. Chile's high voluntary annuitization rate (over 60 percent choosing annuities over programmed withdrawals; Illanes and Padi, 2019) demonstrates that institutional design – including structured choice architecture, regulatory oversight, and limited alternatives – can produce substantially higher take-up than the Anglo-Saxon cases would predict. The Netherlands achieves near-universal annuitization through collective occupational arrangements rather than individual retail choice. These cases complicate any claim that low voluntary annuitization reflects deep and invariant preferences; institutional design, choice architecture, and the available menu of alternatives substantially condition outcomes.

The overall pattern: when annuitization is mandated or embedded in institutional defaults, compliance is high; when the mandate is removed (UK), demand falls substantially; when purely voluntary in a retail context (US, Australia), take-up is low. Whether the low-take-up cases reflect stable preferences, supply-side failures, behavioral frictions, or some combination remains an open question.

6.6 Interpreting the Cross-Cultural Evidence

The cross-cultural record admits three interpretations, each with distinct policy implications. First, behavioral biases (loss aversion, complexity aversion, passivity) may overwhelm rational choice once mandates are removed. The FCA's Retirement Outcomes Review (2018) evidence that many post-2015 decisions reflect passivity rather than deliberate optimization supports this interpretation – large numbers of retirees leaving funds in cash drawdown earning near-zero returns, failing to compare products, or not seeking advice. This interpretive possibility should be taken seriously, as it would imply that removing annuitization mandates reduces welfare for at least some retirees.

Second, the structural channels (pre-existing annuitization, bequest motives, health risk) independently justify low annuitization on rational grounds for the majority of households. When the structural channels predict low annuitization, preference is the more parsimonious explanation.

These interpretations are not mutually exclusive. For retirees who are single, childless, healthy, wealthy, and possess low bequest motives, the structural channels provide weaker justification, and behavioral factors may account for welfare-reducing under-annuitization. This subpopulation is not trivially small (Section 8.2). The aggregate positive finding should be distinguished from the heterogeneous welfare question.

Third, low take-up in the UK, US, and Australia is consistent with the kin-directed investment motives documented in Section 3.3. The annuity, which converts bequeathable wealth into a non-transferable consumption flow, is in tension with the kin-directed preferences evolutionary theory suggests. However, the high-annuitization cases (Chile, Netherlands, Denmark) demonstrate that institutional design can produce high annuitization even where bequest motives are presumably present, suggesting that while bequest motives persistently depress voluntary retail demand, they are not an insuperable barrier when the institutional framework is well designed. The cross-cultural evidence supports a conditional interpretation: low voluntary annuitization reflects the interaction of bequest motives, supply-side frictions, and behavioral factors with institutional design, rather than preferences invariant to institutional context.

7. Synthesis: A Structured Assessment of the Literature

This section moves beyond sequential treatment to assess the literature as an integrated whole: a structured inventory of key contributions (Section 7.1), an analysis of interaction effects (Section 7.2), and a rough decomposition of quantitative importance (Section 7.3).

7.1 A Structured Inventory of the Literature

Table 3 inventories the principal studies, organized by channel, with the final column summarizing the effect on predicted annuitization relative to the Yaari benchmark.

Table 3. Structured Inventory of Key Studies on the Annuity Puzzle

Study	Channel	Method	Key Finding	Effect on Predicted Annuitization
Yaari (1965)	Baseline	Analytical lifecycle model	Under EU, no bequests, and fair pricing, full annuitization is optimal	100% annuitization (theoretical benchmark)
Davidoff, Brown, and Diamond (2005)	Generalization / market completeness	Analytical model with relaxed preference assumptions	Full annuitization holds under weaker utility assumptions but requires complete markets; partial result survives under incomplete markets	100% under complete markets; potentially much lower under incomplete markets
Mitchell, Poterba, Warshawsky, and Brown (1999)	Supply-side pricing	Actuarial valuation of U.S. annuity contracts	Money's worth ratio of 0.80-0.85 for population-average mortality; 0.90-0.94 for annuitant mortality	Reduced; 15-20% load considerably diminishes welfare gains

Study	Channel	Method	Key Finding	Effect on Predicted Annuitization
Mitchell et al. (1999); Finkelstein and Poterba (2002, 2004)	Adverse selection	Actuarial decomposition (Mitchell et al.); empirical analysis of UK market (F&P)	Mitchell et al. estimate adverse selection accounts for approximately 11 cents/dollar (roughly half of total pricing shortfall) for 65-year-old males; F&P confirm selection magnitude is larger in voluntary than compulsory UK markets; F&P find selection on contract structure but not on quantity	Reduced for average-mortality individuals; selection raises effective price
Dushi and Webb (2004)	Pre-existing annuitization	Calibrated lifecycle model with SS and DB pensions	Accounting for pre-existing annuity income, optimal annuitization age ranges from 73 to 82 for couples; some should never annuitize	Sharply reduced; optimal timing delayed or eliminated
Hu and Scott (2007)	Prospect theory	Analytical model under CPT with Tversky-Kahneman parameters	Reservation prices fall below 1 under CPT with baseline parameterization and narrow framing; result sensitive to reference point, loss aversion coefficient (typically 1.5-2.5, though specification-dependent estimates can be higher), and discounting assumptions; qualitative demand suppression robust	Predicted demand zero under baseline narrow-framing parameters; sensitivity to framing and discounting acknowledged
Brown, Kling, Mullainathan, and Wrobel (2008)	Framing effects	Survey experiment	72% prefer annuities in consumption frame vs. 21% in investment frame; 51-percentage-point swing from framing alone	Demand is frame-dependent, violating EU frame invariance; investment framing predicts very low demand

Study	Channel	Method	Key Finding	Effect on Predicted Annuitization
De Nardi, French, and Jones (2010)	Health expenditure risk	Structural lifecycle model with medical expense shocks	Medical expense risk drives elderly precautionary saving; liquid wealth is needed for health shocks	Reduced; precautionary motive conflicts with annuitization
Ameriks, Caplin, Laufer, and Van Nieuwerburgh (2011)	Public care aversion	Strategic survey design separating bequest from care motives	Both bequest motives and public care aversion are economically significant and too prevalent to ignore; Medicaid aversion widespread, particularly among middle class	Reduced; households retain liquid wealth to self-fund LTC and avoid Medicaid
Lockwood (2012)	Bequest motives (calibration)	Calibrated lifecycle model with and without bequests	WTP collapses with bequest motives (25.3% → 3.7% of wealth for fair annuities); predicted ownership falls from 61% to ~10-15% with bequests, to 4-5% with bequests + fixed purchase cost, approximating observed 3.6%	Quantitative match on extensive margin requires bequest motives + purchase frictions; WTP result robust across specifications
Gottlieb (2012, working paper)	Prospect theory (unified model)	Analytical model under CPT	Single CPT framework can account for under-annuitization, guarantee clauses, period-certain features, and joint holding of life insurance and annuities	Zero or near-zero; explains multiple market patterns simultaneously
Pashchenko (2013)	Multiple (pre-annuitization, bequests, housing, minimum purchase)	Structural lifecycle model with sequential decomposition	Four factors matter most: preannuitized wealth, minimum purchase requirement, illiquid housing, bequest motives; full model predicts ~20% participation vs ~5% in data; no single friction sufficient	Substantially reduced but model still overpredicts; highlights importance of joint-channel interaction

Study	Channel	Method	Key Finding	Effect on Predicted Annuitization
Peijnenburg, Nijman, and Werker (2016)	Multiple (incomplete markets, background risk, bequests, default risk)	Calibrated lifecycle model	Full annuitization remains approximately optimal for reasonable parameter values; no single friction examined breaks the full-annuitization result; authors conclude "the annuity puzzle remains a puzzle"	Not reduced within single-model framework for reasonable parameters
Reichling and Smetters (2015)	Stochastic mortality and correlated health costs	Analytical and calibrated model	When mortality is stochastic and correlated with medical expenses, optimal annuity demand can be zero or negative; mechanism operates through correlation structure, not generic medical expense risk	Zero or negative under calibrated health-mortality-cost correlation
Inkmann, Lopes, and Michaelides (2011)	Bequest motives, pre-existing income, preference parameters	Calibrated lifecycle model with UK data	Reasonable preference parameters predict annuity demand levels comparable to the data; bequest motives and pre-existing pension income are key drivers of low demand	Largely explained; reasonable parameters match observed low participation
O'Dea and Sturrock (2023)	Survival pessimism	Calibrated lifecycle model with ELSA subjective beliefs	Individuals underestimate survival by 15-30 pp; pessimism markedly reduces perceived annuity value	Reduced; survival pessimism accounts for significant non-annuitization within EU framework

Two observations emerge. First, independent research traditions – behavioral, structural, health, and insurance economics – converge on the same qualitative finding, indicating

robustness across frameworks. Second, several channel combinations can generate predicted demand at or near zero, though Peijnenburg et al. (2016) find full annuitization approximately optimal within their unified framework.

It is important to note where the literature disagrees, and to do so with specificity. Pashchenko (2013) is particularly relevant because she explicitly nests multiple impediments and performs sequential relative-importance exercises. Her benchmark calibration with all impediments (preannuitized wealth, minimum purchase requirement, illiquid housing, bequest motives) still predicts approximately 20 percent annuity market participation, against roughly 5 percent in the data – a fourfold overprediction. Peijnenburg et al. (2016) reach a similar conclusion, titling their paper 'The Annuity Puzzle Remains a Puzzle.' A specific mechanism in their framework warrants engagement: their model explicitly allows agents to save out of annuity income, rebuilding a liquid wealth buffer after full annuitization. This directly challenges the liquidity argument of Section 4, since the annuity need not irrevocably eliminate access to spendable wealth – in their simulations, agents save approximately 17 percent of annuity income at age 70, accumulating median liquid savings of roughly 25 percent of initial wealth by age 84. However, the practical relevance of this mechanism is conditioned by institutional context. Peijnenburg et al. explicitly assume European conditions under which "out-of-pocket health costs are negligible," and even their own results show notable sensitivity, with the share of liquid financial wealth optimally annuitized dropping to 72 percent for lower-wealth households. The paper also excludes means-tested benefit interactions and minimum purchase requirements. In the U.S. institutional environment, where catastrophic long-term care costs can exceed $100,000 annually and Medicaid spend-down rules penalize accumulated savings, the save-out-of-annuity-income buffer would compete directly with the precautionary demands that Section 4 documents. Inkmann, Lopes, and Michaelides (2011), using UK data, find that reasonable preference parameters can generate annuity demand levels comparable to observed data, with bequest motives and pre-existing pension income as key drivers – leading the authors to question the conventional wisdom that limited participation is a puzzle requiring explanation.

This tension is genuine. The present survey reaches a more decisive conclusion by synthesizing across modeling frameworks: Lockwood (2012) approximates observed ownership

with bequests plus purchase frictions; Reichling and Smetters (2015) generate zero demand through stochastic mortality correlated with health costs; Hu and Scott (2007) generate zero demand under CPT. But Pashchenko's joint model still overpredicts by a factor of four. Two interpretations: the channels she omits (prospect theory, survival pessimism, the full pricing wedge) may account for the residual, or the divergence may reflect differences in calibration and model structure. Pashchenko's residual overprediction and Peijnenburg et al.'s opposing finding represent genuine constraints on the dissolution thesis: they establish that even calibrated multi-channel models leave meaningful uncertainty about optimal annuitization rates. Resolution requires structural models incorporating all channels simultaneously (Section 8.3).

7.2 Interaction Effects: How the Channels Compound

A critical feature of the literature obscured by channel-by-channel treatment is that the explanatory mechanisms plausibly compound, as demonstrated in pairwise cases (e.g., Lockwood, 2012) though not yet tested in a unified model incorporating all channels simultaneously. Pre-existing annuitization through Social Security means the marginal utility of additional annuity income is low; bequest motives mean the cost of annuitization – forfeited bequeathable wealth – is high; health expenditure risk makes liquidity precious; and supply-side frictions render pricing unfavorable. When these channels operate simultaneously, the residual benefit of annuitization may be zero or negative.

The interaction between bequest motives and loading costs illustrates the compounding mechanism. Lockwood (2012) showed that bequests plus purchase frictions reduce predicted ownership from 61 percent to 4-5 percent. At realistic loads of 15-20 percent (Mitchell et al., 1999; confirmed by Wettstein et al., 2021), WTP among consumers with calibrated bequest motives drops to zero. The load eliminates the already-small surplus that remained after accounting for bequests.

A parallel compounding operates between stochastic mortality and bequest motives. Reichling and Smetters (2015) demonstrated that stochastic mortality correlated with medical costs can drive optimal demand to zero even without bequest motives. Adding bequests only strengthens the case, because it introduces an additional cost to a product whose insurance

value has already been undermined. A third compounding occurs between behavioral factors and structural channels: survival pessimism reduces the perceived mortality credit (the primary source of annuity value), while loss aversion increases the perceived cost of premium forfeiture. The channels form layers of resistance, each filtering out additional potential purchasers, and their joint operation is consistent with the low annuitization levels observed. An important qualification: some channels are partially substitutes rather than complements in a unified model. Precautionary savings held against catastrophic health costs occupy the same state-space as wealth that passes as an incidental bequest if the retiree dies healthy (Lockwood, 2018), so incorporating both frictions simultaneously may be sub-additive rather than multiplicative for those specific channels. The compounding characterization is most defensible for channels operating through distinct mechanisms – such as bequest motives and pricing loads, or survival pessimism and narrow framing.

7.3 How Much of the Puzzle Does Each Channel Explain?

Pashchenko (2013) provides the most rigorous benchmark for quantitative assessment, explicitly nesting multiple channels within a single structural model and performing sequential decomposition exercises. Her model incorporates four principal channels: preannuitized wealth (Social Security and DB pensions), a minimum purchase requirement, illiquid housing wealth, and bequest motives. With all four operating jointly, the model predicts approximately 20 percent annuity market participation – against roughly 5 percent in the data, a fourfold overprediction. Importantly, Pashchenko also reports a modified full model that lowers predicted participation to less than 8 percent by making two parameter adjustments within the existing framework: excluding cars and businesses from annuitizable wealth (narrowing the wealth definition) and doubling the minimum purchase requirement from $2,500 to $5,000 in annual annuity income (Pashchenko, 2013, Section 5.4). No new economic channels are introduced; the reduction comes entirely from tightening existing supply-side frictions. This result illustrates how sensitive predicted participation is to institutional parameters, and supports the broader dissolution thesis: even within a single structural framework, realistic calibration of market frictions substantially narrows the gap with observed participation rates.

The channels Pashchenko omits correspond to several of the most powerful demand-suppressing mechanisms identified subsequently. Reichling and Smetters (2015) show that stochastic mortality correlated with medical costs drives optimal annuitization to zero even without bequest motives and under fair pricing – a channel absent from Pashchenko's framework. Hu and Scott (2007) show that under CPT with standard parameters and narrow framing, reservation prices fall below actuarial fair value, predicting zero demand – a preference representation Pashchenko does not consider. O'Dea and Sturrock (2023) show that survival pessimism markedly reduces perceived annuity value within the EU framework itself. Inflation erosion – nominal annuities lose roughly 45 percent of purchasing power by year 20 at 3 percent inflation – is not fully captured in Pashchenko's pricing treatment. Realistic pricing loads (MWR 0.80-0.85) may be quantitatively dominant, eliminating the small surplus remaining after accounting for other channels. Each has been shown individually to generate large demand reductions.

Whether these channels would close Pashchenko's residual gap if incorporated jointly is an open modeling question. The channels are not fully independent: bequest motives, public care aversion, and means-tested transfers affect late-life saving through related marginal utility logic; medical expense risk is often embedded in calibrations used to identify bequest motives; prospect theory replaces rather than supplements the preference representation. Cross-paper synthesis cannot substitute for a structural model incorporating all channels simultaneously. Nevertheless, cross-framework synthesis is appropriate when no single model incorporates all empirically relevant channels, and the logical structure of the argument is stronger than mere qualitative agreement. Multiple channels can each independently generate zero or near-zero predicted demand under standard parameterizations: Reichling and Smetters (2015) under correlated health-mortality risk, Hu and Scott (2007) under CPT with narrow framing, Lockwood (2012) with calibrated bequests plus realistic loads. If individual channels suffice to eliminate demand in isolation, their joint operation under any reasonable structural specification must also predict low demand – the convergence is overdetermined, though the sub-additivity noted in Section 7.2 means the quantitative magnitude under joint estimation remains an open question. No framework predicts high voluntary immediate-annuity demand once realistic institutional features are incorporated, and the specific channels Pashchenko

omits are precisely those the subsequent literature identifies as most powerful. The aggregate puzzle is substantially attenuated and likely dissolved for most households under realistic joint parameterizations, though a unified structural model incorporating all channels remains the necessary confirmatory test (Section 8.3). This dissolution claim applies specifically to classic immediate life annuities; for deferred income annuities, where structural channels are substantially attenuated, behavioral factors bear the primary explanatory load and the puzzle remains genuinely open (Section 8.1). The aggregate finding should be distinguished from the heterogeneous normative question addressed in Section 8.2.

8. Remaining Questions and Directions for Future Research

The dissolution of the aggregate puzzle does not mean that all questions about annuitization have been answered.

8.1 The Deferred Income Annuity Puzzle

The dissolution argument applies most directly to classic immediate life annuities. The strongest residual challenge concerns deferred income annuities (DIAs), including qualified longevity annuity contracts (QLACs). These products address several of the most powerful objections: the consumer retains full access to wealth until a specified advanced age, typically 80 or 85, preserving liquidity during the period when health expenditure risk is most acute, while providing longevity insurance precisely at the ages when outliving resources is most threatening. The premium is substantially smaller as a fraction of total wealth than for an immediate annuity, reducing the bequest sacrifice. The SECURE 2.0 Act of 2022 raised the per-person QLAC contribution limit to $200,000 (indexed for inflation; $210,000 for 2025-2026), removing the previous 25-percent-of-balance constraint.

Yet demand remains near zero. This fact warrants careful analysis because the structural channels documented in Sections 3-6 apply with considerably reduced force to DIAs. Pre-existing annuitization is less relevant because DIA payouts would supplement, not duplicate, Social Security income at advanced ages. Bequest motives apply with reduced force because the DIA premium is a modest fraction of total wealth. Health expenditure risk is directly

addressed by the product's design: liquidity is preserved during the years when catastrophic health costs are most likely. Pricing remains a friction – Wettstein et al. (2021) report DIA money's worth ratios around 50 cents per dollar of premium in nominal terms, though the insurance value on a utility basis is substantially higher – but the QLAC's location within a tax-advantaged retirement account mitigates some of the pricing objection. The $210,000 cap itself limits the product for upper-wealth households.

One structural friction does apply with full force to DIAs: inflation risk. Because payouts are deferred 15-20 years, a nominal DIA is massively exposed to purchasing-power erosion between purchase and payout commencement – at 3 percent inflation, a nominal DIA purchased at 65 with payouts beginning at 85 would have already lost roughly 45 percent of its real value before the first payment is received. This makes the product objectively riskier from a real-consumption perspective and compounds behavioral aversion. Interest rate lock-in compounds this friction: because DIA payouts are fixed at purchase, consumers face an option value of waiting for higher rates, suppressing current demand.

The most plausible explanations for near-zero DIA demand are behavioral, operating through several reinforcing channels whose relative importance differs from the SPIA case. Loss aversion operates against a product that requires forfeiting current wealth for payments contingent on survival to age 85 – the "loss" (premium paid) is certain and immediate, while the "gain" (income stream) is contingent and remote. Survival pessimism (O'Dea and Sturrock, 2023) reduces the perceived value of a product whose entire benefit is conditional on reaching advanced age; given that individuals underestimate survival probabilities by 15-30 percentage points, the subjective probability of collecting DIA benefits is considerably lower than the actuarial probability. The cognitive challenge is substantial: evaluating a 20-year irreversible commitment based on survival probabilities that individuals systematically underestimate. Distribution channel frictions compound the behavioral obstacles: financial advisors compensated through assets under management have little incentive to recommend a product that removes assets from the fee-generating portfolio, while the modest commissions on QLACs relative to indexed annuities create weak incentives for commission-based advisors.

The relative importance of these channels likely differs between SPIAs and DIAs in testable ways. For SPIAs, structural channels (pre-existing annuitization, bequest motives,

health-expenditure risk) do most of the explanatory work, with behavioral factors playing a confirmatory role. For DIAs, structural channels are substantially attenuated, implying that behavioral and distributional factors must bear the primary explanatory load. An empirical test that could discriminate: if DIA take-up increases substantially under automatic enrollment or simplified choice architecture – as 401(k) participation did under auto-enrollment (Madrian and Shea, 2001) – but SPIA take-up does not, this would confirm that behavioral rather than structural frictions are the binding constraint for deferred products. The DIA puzzle is the most plausible candidate for a genuine residual anomaly and the strongest test case for whether behavioral biases constitute a policy-relevant welfare problem rather than a confirmatory gloss on structurally justified behavior.

8.2 Heterogeneity in Optimal Annuitization

The dissolution of the aggregate puzzle does not imply universal welfare optimality. The subpopulation for whom the welfare case is strongest includes: single individuals without children (approximately 16 percent of Americans aged 65-74 are childless; U.S. Census Bureau, 2021); those in good health with above-median expected longevity; those with financial wealth in the top decile, where pre-existing annuitization through Social Security may cover only one-third of total wealth (Lockwood, 2012); and those reporting low bequest motives (approximately 25 percent of elderly singles are estimated to have no bequest motive; Kopczuk and Lupton, 2007). The intersection is not trivially small: even a conservative estimate suggests potentially millions of American households for whom the structural case against annuitization is weaker than for the typical retiree, and for whom welfare-improving annuitization may be achievable through product reform – better pricing, inflation protection, or institutional pooling.

Future research should quantify the welfare gains from annuitization that would emerge under improved pricing and product design for these households, using unified structural models that incorporate all relevant channels simultaneously.

8.3 Directions for Future Research

Five directions merit emphasis. First, structural models jointly incorporating pre-existing annuitization, bequest motives, health risk, stochastic mortality, and realistic pricing are needed. The divergence between this survey's synthesis and Peijnenburg et al. (2016) can only be resolved by models allowing all channels to operate simultaneously, quantifying whether the interaction effects hypothesized in Section 7.2 hold under joint estimation. Part of the reason this has not been done is computational: solving a lifecycle model with bequest motives, stochastic health costs, stochastic mortality, realistic pricing, and CPT preferences involves a high-dimensional state space that remains at the frontier of current methods.

Second, DIA and QLAC demand warrants specific analysis as the strongest test of whether behavioral biases generate welfare losses in the annuitization domain.

Third, the optimal default literature (Carroll et al., 2009; Beshears et al., 2014) has direct implications. If the UK post-2015 experience reflects default effects rather than informed preference, optimal default design for retirement income may be more productive than information provision or tax incentives.

Fourth, modern longevity-pooling designs warrant investigation as potential alternatives to traditional annuitization. Milevsky and Salisbury (2015) analyze optimal retirement income tontines – pooled structures that distribute mortality credits without insurer guarantees, thereby eliminating counterparty risk and potentially reducing costs. Because tontine participants retain a claim on a shared pool rather than purchasing an irrevocable insurer promise, such designs may partially address the liquidity, bequest, and counterparty-risk objections that suppress demand for conventional annuities, though they introduce payout volatility and regulatory challenges that remain unresolved.

Fifth, Social Security delay deserves investigation as arguably the highest-value annuitization mechanism available to many American retirees, particularly married primary earners with liquid assets to bridge the delay period – though households facing health constraints, work limitations, or inadequate liquid assets may be unable to delay. Delaying benefits from age 62 to 70 increases benefits through both the elimination of early-claiming reductions (ages 62-66/67) and delayed retirement credits of approximately 8 percent per year (ages 66/67-70,

after which no further credits accrue) on an inflation-indexed, survivor-benefit-eligible annuity backed by the federal government – substantially better pricing than any private product. For households with sufficient bridging resources and above-average life expectancy, delay is the most actuarially favorable annuitization available, reinforcing the pre-existing annuitization channel documented in Section 3.1. The actuarial fairness of delay is heterogeneous by socioeconomic status, and many households lack liquid wealth to finance the foregone benefits, but the existence of a superior public annuitization mechanism underscores that the relevant policy question is product-specific rather than categorical.

8.4 Limitations

Several limitations should be acknowledged. First, the cross-cultural evidence does not permit clean causal identification: the six institutional contexts examined involve multiple simultaneous differences, and the high-annuitization cases (Chile, Netherlands) demonstrate that institutional design can produce outcomes substantially different from the low-take-up pattern. Second, the quantitative match between Lockwood's (2012) calibration and observed ownership rates depends on specific bequest parameterizations that, while empirically defensible, are not the only defensible calibrations. Third, the behavioral evidence (Sections 5.1-5.5) documents systematic departures from expected utility that, while confirming the positive prediction of low annuitization, may simultaneously identify welfare-reducing biases for specific subpopulations – a possibility that the aggregate dissolution thesis should not be read as dismissing. Fourth, the pre-existing annuitization argument (50-75 percent of retiree wealth in annuitized form) relies primarily on cohorts with substantial DB pension coverage; as the long-run shift from DB to DC continues, the annuitized share for future retirees will decline substantially, resting principally on Social Security alone. For cohorts entering retirement with near-zero private DB coverage, the aggregate puzzle may re-emerge as a genuine normative challenge even if it is currently dissolved for cohorts with substantial pre-annuitization. Fifth, the welfare conclusions are framework-dependent: this survey implicitly adopts expected utility with calibrated parameters as the normative benchmark, but alternative behavioral welfare frameworks (Bernheim and Rangel, 2009) may yield different assessments, particularly for the subpopulation identified in Section 8.2.

9. Conclusion

The annuity puzzle has occupied household finance researchers for six decades. The accumulated evidence supports the conclusion that the aggregate puzzle – the gap between Yaari's (1965) full-annuitization prediction and the observed 3-6 percent ownership rate for classic immediate life annuities – is better understood as a limitation of the benchmark model than as an anomaly in consumer behavior. The Yaari result assumes a consumer who holds no bequest motive, faces no health shocks, inhabits a frictionless market, and processes decisions with the frame-invariance of the expected utility axioms.

"Dissolving" the puzzle means demoting it from a first-order anomaly in consumer behavior to a predictable consequence of the benchmark's restrictive assumptions – not claiming that all questions about annuitization are settled. The case for retiring the Yaari-benchmark puzzle framing operates through three mutually reinforcing layers. First, the Yaari result rests on nine restrictive assumptions, and under cumulative prospect theory with standard parameters, predicted demand falls to zero (Hu and Scott, 2007). Second, corrected inputs converge on observed levels: pre-existing annuitization places many current retirees at 50-75 percent annuitized, depending on wealth and pension coverage (Dushi and Webb, 2004), bequest motives combined with purchase frictions reduce predicted ownership to approximately 4-5 percent (Lockwood, 2012), and health expenditure risk correlated with stochastic mortality can drive demand to zero (Reichling and Smetters, 2015). Third, supply-side frictions render real products far less attractive than their theoretical counterparts (Mitchell et al., 1999; Finkelstein and Poterba, 2002).

Not all researchers share this assessment. Pashchenko's (2013) structural model incorporating multiple channels still overpredicts participation by a factor of four, and Peijnenburg et al. (2016) find full annuitization approximately optimal in their calibrated framework, titling their paper 'The Annuity Puzzle Remains a Puzzle.' The divergence reflects genuine methodological disagreement about whether channels should be evaluated within unified models or synthesized across frameworks; harmonization remains an important open task.

Retiring the puzzle framing does not mean retiring the research program – and the ongoing shift from defined-benefit to defined-contribution pensions will reduce pre-existing

annuitization for future cohorts, potentially restoring the normative force of the annuitization question even as the positive puzzle dissolves for current retirees. The deferred income annuity puzzle represents the strongest residual test case: a product that addresses the liquidity and health-risk objections and yet attracts near-zero demand. The dissolution of the aggregate puzzle does not imply that annuities are valueless, nor that all households are making welfare-optimal decisions. Partial annuitization would be welfare-improving for some households – particularly those who are single, healthy, wealthy, and possess low bequest motives – provided that pricing, inflation protection, or institutional design improve beyond current retail market conditions. Moreover, low annuitization can be simultaneously individually rational given current products and evidence of welfare losses from insurance-market incompleteness.

The relevant questions going forward include: for which households would additional annuitization improve welfare? What default and menu design in DC decumulation would best serve heterogeneous retirees? Can institutional pooling or public provision offer higher-money's-worth longevity insurance than retail markets? How does the annuitization decision interact with LTC risk and Medicaid design? These questions direct research toward populations, products, and institutional arrangements where the gap between optimal and actual annuitization is largest – an agenda the "retire the puzzle" framing is designed to facilitate, not foreclose.

The broader methodological lesson may be the most important. When "puzzle" becomes the organizing concept for a research program, it can channel inquiry toward explaining away observed behavior rather than questioning the benchmark – a dynamic not unique to annuities but observable wherever knife-edge theoretical predictions are treated as robust empirical expectations. When a theoretical prediction requires a conjunction of nine counterfactual assumptions, the failure of the prediction need not be regarded as a puzzle – it is a predictable consequence of the benchmark's limitations.

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